Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 33: Too Many for the Boxes

The Pigeonhole Principle

Chartrand & Zhang, Section 8.3, pp. 286–292

Name: _____ Date: _____

You met this in Week 7

Result (5.3) said: if $f : A \to B$ is one-to-one, then $|B| \geq |A|$. Its contrapositive: if $|B| < |A|$, then no function $A \to B$ is one-to-one — some two elements of A must share an image.
That contrapositive is today's principle. It has a name, a long history, and a remarkable number of uses for something so obvious.

By the end of today you should be able to

1. State the pigeonhole principle, including the $\lceil n/k \rceil$ form.
2. Identify the pigeons and the holes in a problem.
3. Explain why a pigeonhole argument proves existence without naming the object.

1. The principle

The Pigeonhole Principle

If a set with n elements is divided into k pairwise disjoint subsets, then at least one subset has at least $\lceil n/k \rceil$ elements.

Here $\lceil x \rceil$ is the **ceiling** — the smallest integer greater than or equal to x. So $\lceil 12.5 \rceil = 13$ and $\lceil 4 \rceil = 4$.

The simplest case: n pigeons in $k < n$ holes means some hole holds at least 2.

Example 8.20. A desk has 8 pigeonholes. File 100 papers, and some hole holds at least $\lceil 100/8 \rceil = 13$.

2. The proof is by contradiction

Proof. Assume, to the contrary, that every subset has fewer than $\lceil n/k \rceil$ elements. Since $\lceil n/k \rceil$ is an integer, each has at most $\lceil n/k \rceil - 1$ elements, which is $\leq n/k$. The subsets are disjoint and cover the set, so

$$n = |S_1| + |S_2| + \cdots + |S_k| < k \cdot \frac{n}{k} = n,$$

giving $n < n$ — a contradiction. ■

That is Lecture 27's method, and the addition step uses the Addition Principle from Monday.

3. Name the pigeons and the holes

Most pigeonhole problems are hard only until you decide what the pigeons and holes are. Then they are one line.

Claim	Pigeons	Holes	Why
Among 13 people, two share a birth month.	people	12 months	$13 > 12$
Among 11 integers, two share a last digit.	integers	10 digits	$11 > 10$
Among 15 days, three fall on the same weekday.	days	7 weekdays	$\lceil 15/7 \rceil = 3$

How many are needed?

To guarantee that some hole holds at least r items when there are k holes, you need

$$N = k(r-1) + 1$$

items. With $k(r-1)$ items, you could put exactly $r-1$ in every hole.

4. A cleverer choice of holes

Claim

If $n+1$ integers are chosen from $\{1, 2, \ldots, 2n\}$, then two of them are consecutive.

The trick is choosing the holes. Use the n pairs

$$\{1,2\},\ \{3,4\},\ \ldots,\ \{2n-1,\ 2n\}.$$

These are n holes, and we have $n+1$ pigeons, so two chosen integers land in the same pair — and the two members of a pair are consecutive. ■

Sharpness. With only n integers it can fail: $\{1, 3, 5, \ldots, 2n-1\}$ has no two consecutive.

The holes are yours to choose

Nothing in the problem statement mentions pairs. Inventing the right holes *is* the proof, and it takes practice to see. The standard move: group the possible values so that any two items in one group have the property you want.

5. Existence without a witness

A pigeonhole argument proves that two people share a birth month without telling you which two.

Lecture 26, again

That is a **non-constructive existence proof**, like Result 3.38. The pigeonhole principle is one of the most common sources of them.
A famous example (*MCS* Section 14.8): Boston has about 500,000 non-bald people, each with at most 200,000 hairs. So at least three people in Boston have *exactly* the same number of hairs on their heads. Nobody knows who they are. They must exist.

Exit ticket

1. How many integers must you choose to guarantee two leave the same remainder on division by 5? Name the pigeons and holes.

2. Show that any 6 integers chosen from $\{1, \ldots, 10\}$ include two summing to 11. (Hint: which pairs sum to 11?)

Before Monday

Read Section 8.4, "Permutations and Combinations," pp. 292–302.

Self-check (not collected). Section 8.3, Exercises 1, 3, and 5 — odd-numbered, answers in the back.

Coming up

Homework 11 is due tonight, Friday, November 6, 11:59 p.m.
Quiz 10 is Monday, November 9 — Sections 8.1 and 8.2.
Homework 12 covers Sections 8.1–8.3 and is due Friday, November 13.

Questions I still have